

Data Science

Class 1

Core Algorithms

Regression — Linear Reg, Logistic Reg.
 Classification — Decision Trees, K-Nearest Neighbors, SVM, Random Forest, Ensemble Tech.
 Clustering — K-Means Clustering, DBSCAN
 Reinforcement Learning

Lab work

- ① Assignments
- ② Final Project

Process Basics

- Numpy, pandas, Scikit Learn
- Data Exploration
- Data Cleaning
- Model Evaluation

Problem Statement

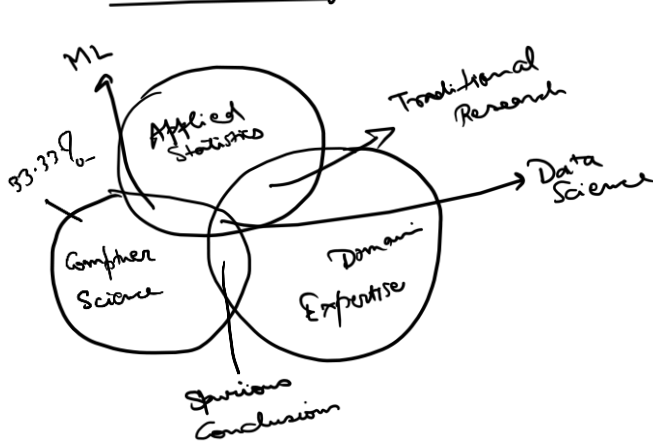
Figure out whether a candidate will default on their loan or not.

Variables	Income	# of cars	Education Level	Loan Amt.	Residency Type	Default (Y/N)
→	50L	2	M.Tech	—	3 BHK	N
→	—	—	—	—	—	N
→	—	—	—	—	—	N
→	—	—	—	—	—	N
→	—	—	—	—	—	Y
→	—	—	—	—	—	Y
→	—	—	—	—	—	N
→	—	—	—	—	—	Y
→	70L	1	B.Tech	1cr	2BHK	N

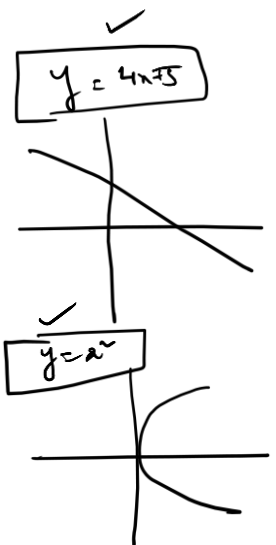
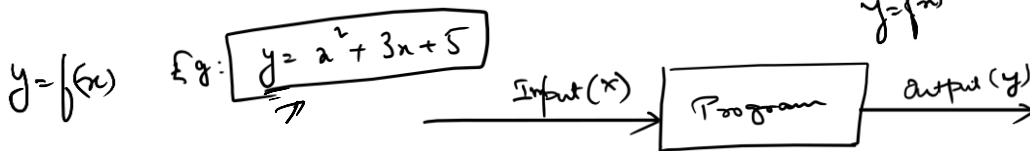
Independent Variables

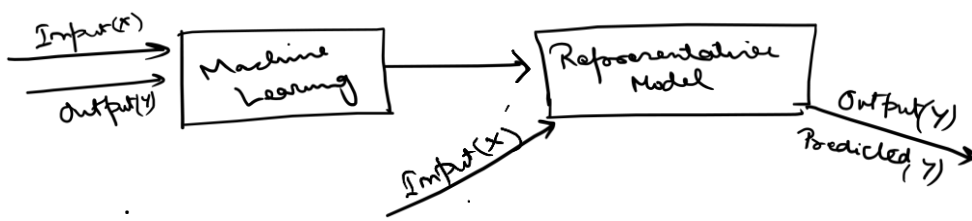
Target Variable.

Income & Default → choosing the variables is called Domain Knowledge.

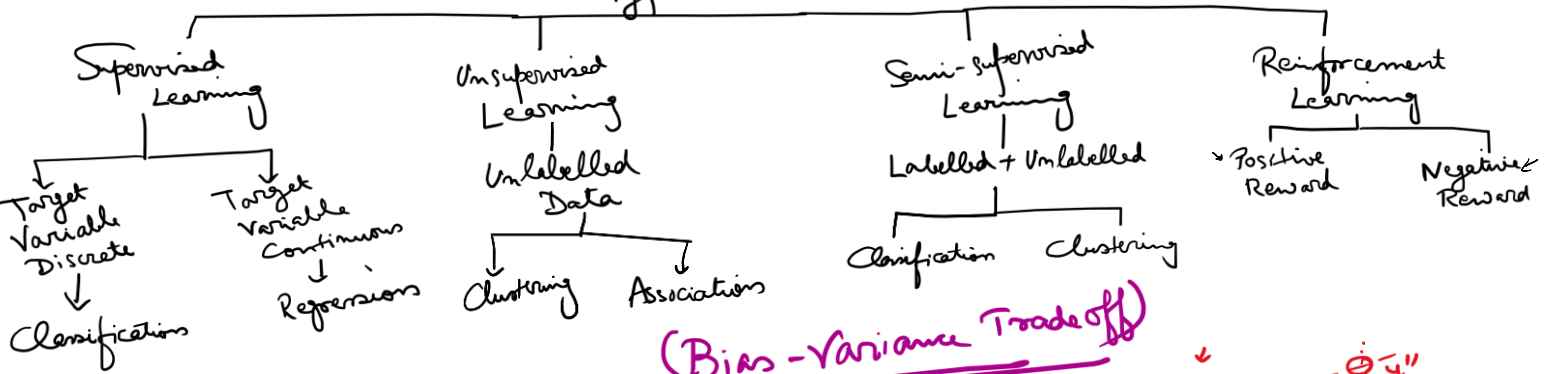


Data Science

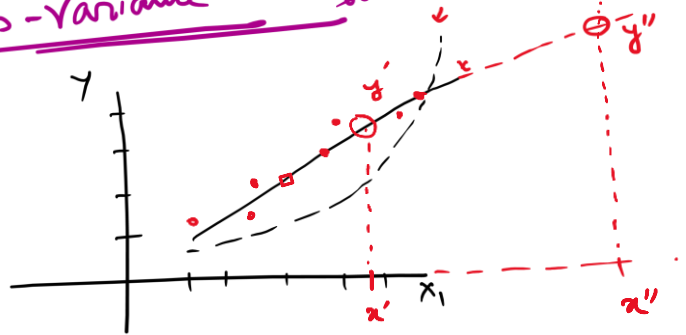




Machine Learning Types



(Bias-Variance Tradeoff)



Regression

Stock Value (X1)	Demand (X2)
x11	x21
x12	x22
x13	x23
...	...

Price (y)

y1
y2
y3
...

- Linear Regression
- Logistic Regression

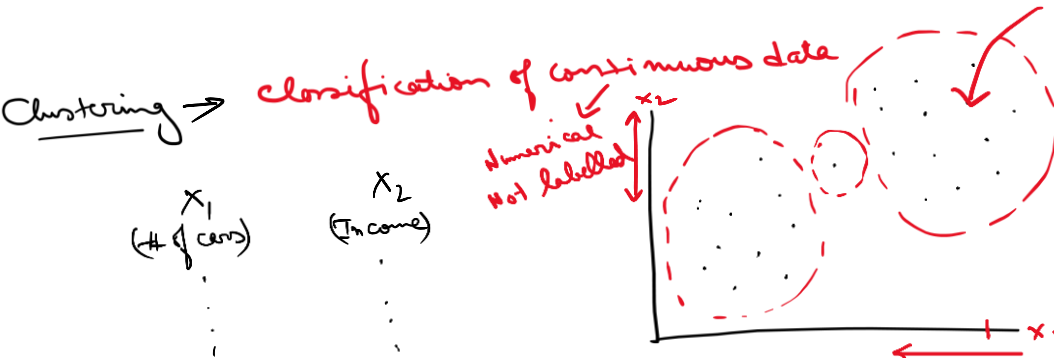
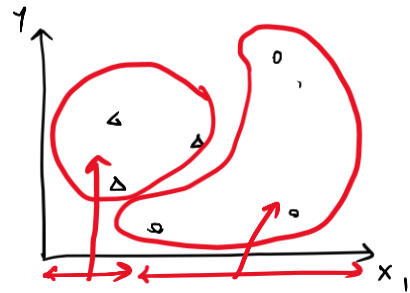
Is the target variable numerical / categorical?

Δ - Yes
o - No

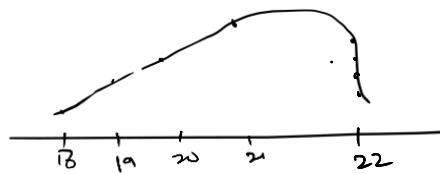
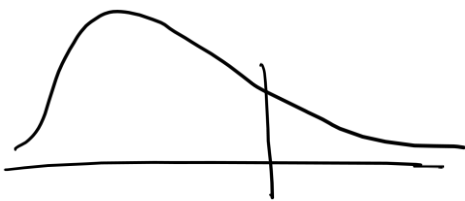
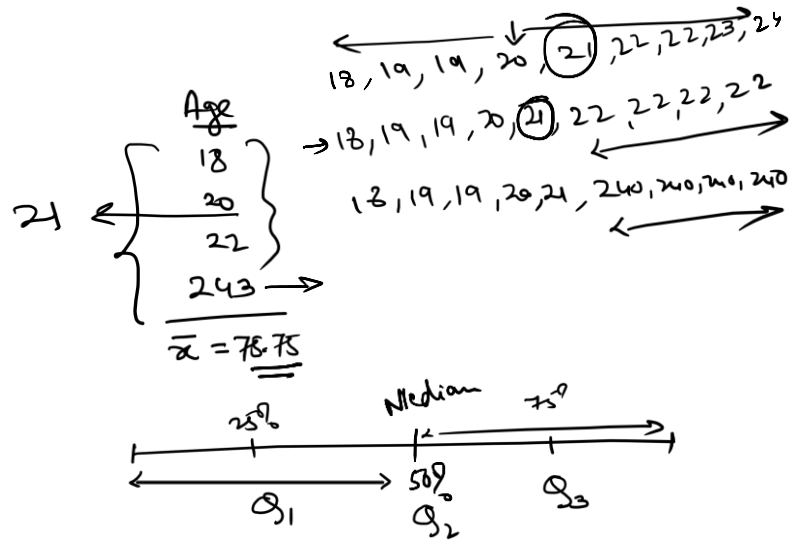
Classification

(# cars) X1	(Income) X2	(Default) Z
3	58L	0
2	28L	1
0	13L	0
1	10L	1
...	...	...

Yes = 1
No = 0



- Customer Segmentation
- Building marketing Strategies



array - homogeneous

series = {key1:[], key2:[]}

```

graph TD
    Root[ ] --- Delete[Delete]
    Root --- Replace[Replace]
    Replace --- Continuous[Continuous variables  
(median)]
    Replace --- Categorical[Categorical variables  
(mode)]

```

Outliers are detected in

2 ways :- Box Plot / IQR

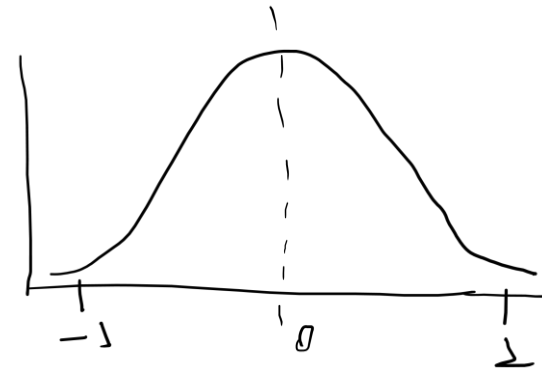
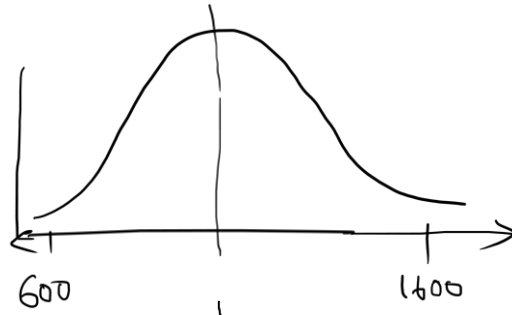
$$\text{Max} = Q_3 + 1518R$$

$$P_m = Q_1 - 1.5 I Q_2$$

99% of the data should be b/w
[min, max]

(II) Z-Score = $Z_i = \frac{x_i - \bar{x}}{s.d(x)} \rightarrow \text{Mean}$

$\frac{x}{x_1}$	$\frac{z}{z_1}$
x_2	z_2
x_3	z_3
x_4	
$\vdots$	$\vdots$
x_n	z_n



After this transformation, 95% of the data (Z) should lie b/w $(-1, 1)$

outlier = $Z > 3$ or $Z < -3$

99.5% of the data will lie b/w $[Z-3, Z+3]$

Z-scoring $\leftrightarrow$ Normalization $\leftrightarrow$ Standardization

Treating categorical variables; - Label Encoding
- One Hot Encoding

Score	Score (LE)	Score Low	Score Med	Score High
Low	0	1	0	0
Low	0	1	0	0
Med	1	0	1	0
High	2	0	0	1
Med	1	0	1	0

Disadvantage (LE) - Misinterpretation of ordinal values
- Arbitrary numerical system
- Complex if too many categories

Disadvantage (OAE) - Too many new variables created

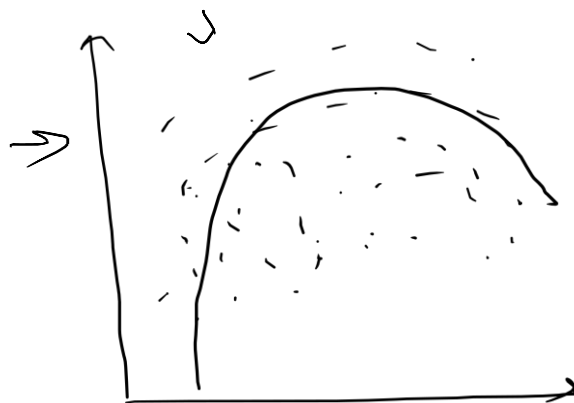
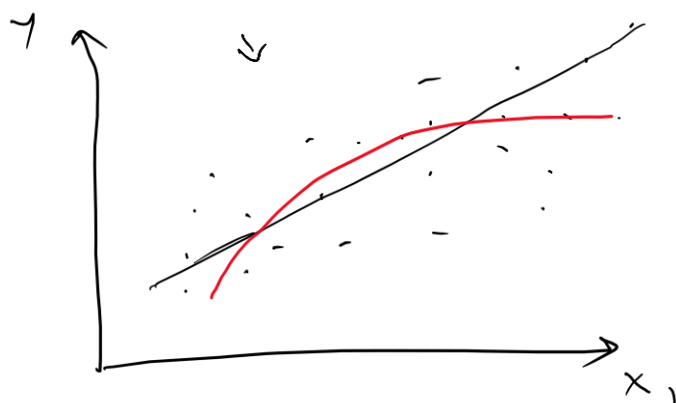
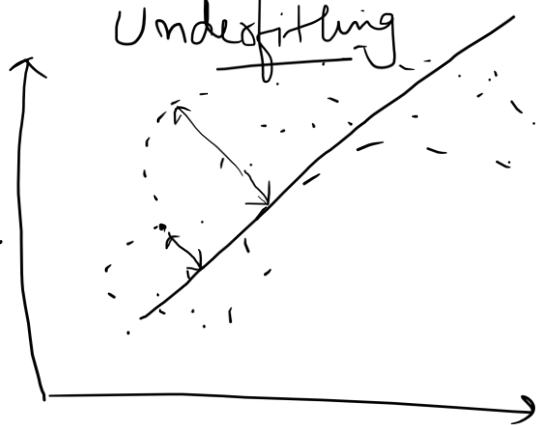
$$y = \beta_0 + \beta_1 x_1$$

Bias

Overfitting

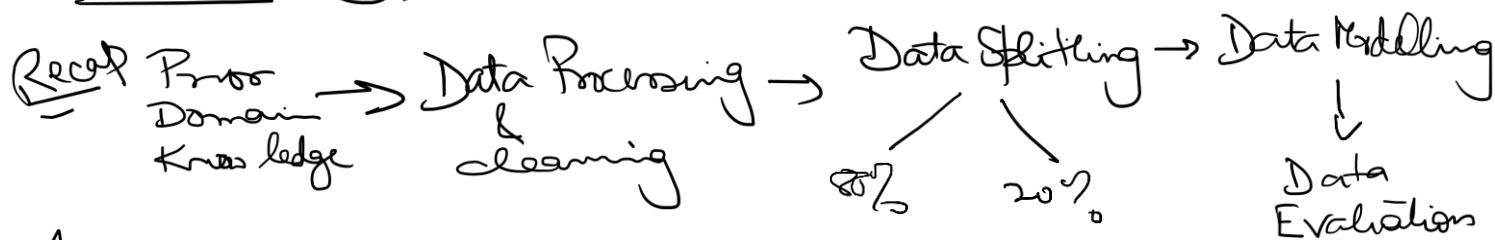
Var

Underfitting



Bias-Variance Tradeoff

Linear relationship b/w two variables
Linear Regression → It predicts the cont. data in target variable



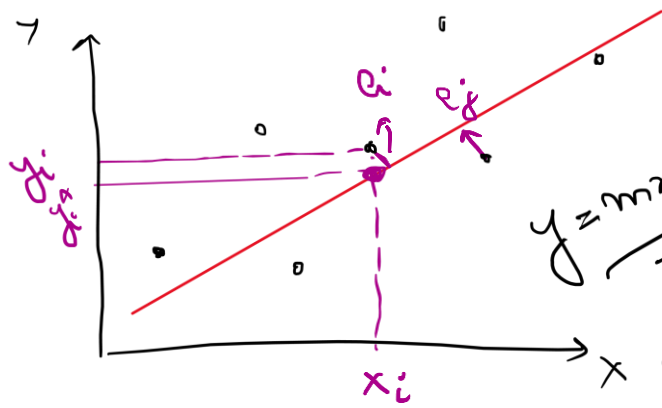
Actual y Predicted $\hat{y}$

$- y_1 \longleftrightarrow \hat{y}_1$
 $- y_2 \longleftrightarrow \hat{y}_2$
 $- y_3 \longleftrightarrow \hat{y}_3$

x
 x_1
 x_2
 x_3
 $\vdots$

$y(\text{Target})$
 y
 y_1
 y_2
 y_3
 $\vdots$

$(y_i - \hat{y}_i)$
Residual
 $y_1 - \hat{y}_1$
 $y_2 - \hat{y}_2$
 $y_3 - \hat{y}_3$
 $\vdots$



→ Dist. of the best regression line from the actual target variable.

$$y = mx + c$$

$$y = \beta_0 + \beta_1 x$$

$$\hat{y} = \beta_0 + \beta_1 \hat{x}$$

Target → Minimize the sum of squares of the residuals.

$$\sum_{i=1}^n e_i^2 \text{ is to be minimized}$$

$$\frac{\partial \sum e_i^2}{\partial \beta} \bigg|_{(\hat{\beta}_1, \hat{\beta}_2)} = 0$$

$$(y_i - \hat{y}_i)^2 \Rightarrow (\hat{\beta}_0 + \beta_1 \hat{x}_{1i} - \beta_0 - \beta_1 \hat{x}_{1i})^2$$

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i}$$

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 \hat{x}_{1i} + \hat{\beta}_2 \hat{x}_{2i}$$

$$\beta_1 (x_{1i} - \hat{x}_{1i}) + \beta_2 (x_{2i} - \hat{x}_{2i})$$

$$\text{Best SLR model line} = \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Bias

Var

Errors not explained by trained model
→ Errors in training

Errors not explained by test data
→ errors while pred.

$$\frac{\partial}{\partial \beta} \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = 0$$

$$\hat{\beta}_1 = \frac{\left(\sum xy - \frac{\sum x \sum y}{n} \right)}{\left(\sum x^2 - \frac{(\sum x)^2}{n} \right)}$$

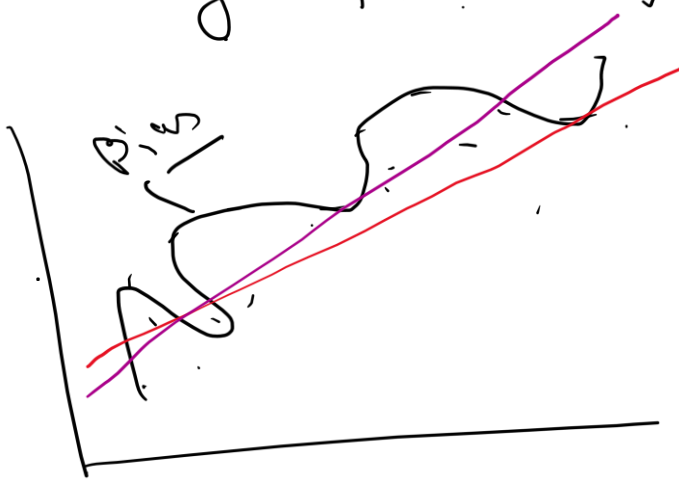
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$\text{Cov}(x, y)$

$\text{var}(x)$

X	y
train	
Test	

$$e_i = y_i - \hat{y}_i$$



Ordinary Least Square

Biased $\rightarrow$ Var $\uparrow$
var $\downarrow \rightarrow$ Bias $\uparrow$

SLR $\rightarrow$ fitted using OLS
 $\Downarrow$
minimizes RSS

Basic Assumption of a SLR model:

- $\rightarrow y_i$ (target) $\rightarrow$ continuous
- $\rightarrow (x_1, x_2) \rightarrow$ Pairwise independent.
- $\rightarrow E_i \rightarrow$ Homoskedastic - variance $\rightarrow$ should be constant

$$y = \beta_0 + \beta_1 x + \underline{\underline{E_i}}$$

$\rightarrow (x_i, y_i) \rightarrow$ should have a linear relationship

Recap → ① What is SLR?

② Assumptions of SLR?

③ Fitting SLR using OLS method

④ Prediction using fitted SLR model

$$y = \beta_0 + \beta_1 x \leftarrow \text{SLR}$$

→ Predict the target variable using multiple predictors.

$$\tilde{y} = \beta_0 + \beta_1 \tilde{x}_1 + \beta_2 \tilde{x}_2 + \dots + \beta_p \tilde{x}_p \leftarrow \text{MLR model}$$

Linear Algebra → n obs. → p indep. var.

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix}_{n \times 1} = \begin{bmatrix} \beta_0 + \beta_1 x_{11} + \beta_2 x_{21} + \dots + \beta_p x_{p1} \\ \beta_0 + \beta_1 x_{12} + \beta_2 x_{22} + \dots + \beta_p x_{p2} \\ \vdots \\ \beta_0 + \beta_1 x_{1n} + \beta_2 x_{2n} + \dots + \beta_p x_{pn} \end{bmatrix}$$

$$\tilde{y} = \begin{bmatrix} 1 & \overbrace{x_{11}}^{x_1} & \overbrace{x_{21}}^{x_2} & \dots & \overbrace{x_{p1}}^{x_p} \\ 1 & x_{12} & x_{22} & \dots & x_{p2} \\ 1 & \vdots & \vdots & \dots & \vdots \\ 1 & x_{1n} & x_{2n} & \dots & x_{pn} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}$$

$$\tilde{y} = X \tilde{\beta} + \tilde{\epsilon} \leftarrow \begin{matrix} \text{residual} \\ \text{bias} \\ \text{error} \end{matrix}$$

To fit MLR model :- Gauss-Markov Theorem

→ To minimize Residual Sum of Squares (RSS)

$$\begin{aligned}
 \text{RSS} &= \sum_{i=1}^n \tilde{e}_i^2 \\
 &= \tilde{e}' \tilde{e} \\
 &= \left. \frac{\partial \tilde{e}' \tilde{e}}{\partial \tilde{\beta}} \right|_{\tilde{\beta} = \hat{\tilde{\beta}}} = 0
 \end{aligned}$$

$\begin{bmatrix} e_1 & e_2 & e_3 & \dots & e_n \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}$

$$\hat{\tilde{\beta}} = (X'X)^{-1} X'y \leftarrow \text{Generalized Least Square Method (GLS)}$$

Assumptions

- ① Target variable (y) must be normally distributed
- ② (X_i, X_j) must be pairwise independent and there must not be any multicollinearity
- ③ All e_i must be independent of each other and have the same variance (spread) with mean 0

homoskedastic

$$\textcircled{4} \tilde{e} \sim N(0, \Sigma) \quad \left| \quad \Sigma = \begin{bmatrix} \sigma^2 & 0 & 0 & 0 & \dots & 0 \\ 0 & \sigma^2 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & \dots & \dots & \sigma^2 \end{bmatrix} \right.$$

Effects of multicollinearity

$x_1, x_2, \dots, x_p \rightarrow x_3, x_4$

Variance inflation $\rightarrow$ cause extra bias towards the impact

Impact of other imp variables will get masked/reduced $\rightarrow$ of x_3 & x_4

Preprocessing

Solution: $VIF \rightarrow$ Variance Inflation Factor

noted $\rightarrow x_3 \rightarrow VIF \rightarrow > 5 \rightarrow$ Drop the variable with highest VIF score
 $\rightarrow$ We have $(p-1)$ variables

$x_4, x_5, \dots$
 $\downarrow \quad \downarrow$
3.7 1.2
 $\downarrow \quad \downarrow$
2.7 0.56

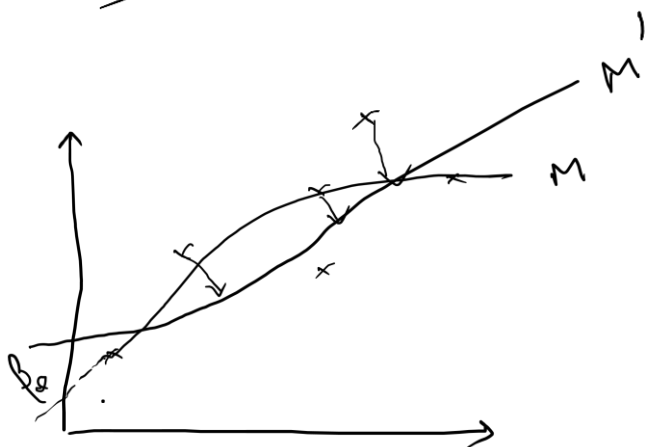
MLR evaluation $\rightarrow$ SLR $\rightarrow R^2, MAE, MSE, Adj R^2$

$$\underline{Adj R^2} = 1 - \frac{(1 - R^2) \cdot \frac{n-1}{n-p-1}}$$

$\rightarrow$ Why $adj R^2$ instead of just R^2 ?

$\downarrow$
 $p \uparrow$; normal $R^2 \uparrow$.

H.W $\rightarrow$ 5 ways to reduce underfitting in MLR?
 $\rightarrow$ 5 ways to reduce overfitting in MLR?



$M' \rightarrow$ Penalty to M

$\downarrow$
new RSS is minimized.

Penalty $\rightarrow$ over the model parameters

Ridge Regression

$$RSS' = RSS + \lambda \sum_{i=1}^p \beta_i^2$$

$$\lambda \in (0, \infty)$$

Lasso Regression

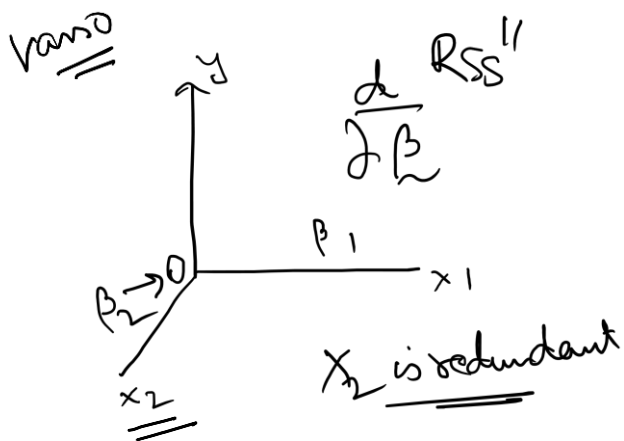
$$RSS'' = RSS + \lambda \sum_{i=1}^p |\beta_i|$$

Penalty multiplier

Diff b/w Ridge & Lasso Regression:-

$\sum \beta_i \rightarrow \beta_i$ of $X_i \rightarrow$ impact will be magnified and it will use all X_i 's as important

~~Ridge~~



If $\lambda < 1$
 $|\beta_i| \rightarrow 0$, help in identifying redundant variable as $\beta_i \rightarrow 0$

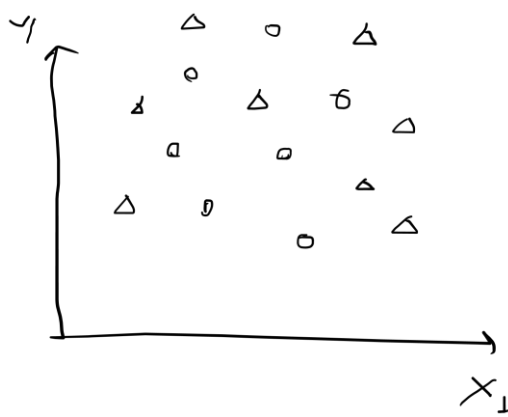
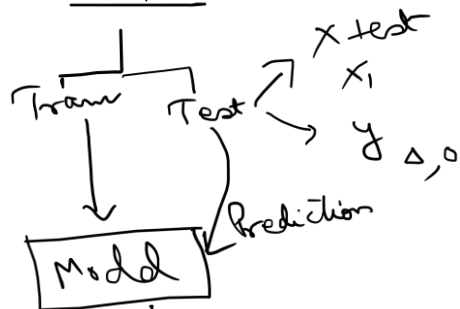
find λ where $adj R^2$ is max.

Q) Which is better Ridge or Lasso? How to choose b/w Ridge & Lasso? $\leftarrow$ H.W.

Classification Day 7

Confusion Matrix

Data Split



Predicted class, compared with actual test class

<u>y_pred</u>	<u>y_test (actual)</u>
Δ	$\Delta \checkmark$
Δ	$\Delta \checkmark$
$\circ$	$\Delta \leftarrow$ mis classification
$\circ$	$\circ \checkmark$
Δ	$\circ \leftarrow$ mis
$\circ$	$\circ \checkmark$
Δ	$\Delta \checkmark$
$\circ$	$\Delta \leftarrow$ mis
$\circ$	$\circ \checkmark$
Δ	$\Delta \checkmark$

n obs.

Actual

	Δ	$\circ$
Δ	True Positive TP (4)	False Positive (1) FP
$\circ$	False Negative (2) FN	True Negative (3) TN

Annotations

Δ - Positive
 $\circ$ - Negative

How to evaluate a trained Class Model?

$$\textcircled{1} \underline{\text{Accuracy}} = \frac{TP + TN}{(TP + FP + FN + TN)} = \frac{7}{10} = 70\%$$

Detection of cancer in 100 patients (testing set)

Cancer - 1 (+), T
No cancer - 0 (-), F

Actual

1 0

Pred	1	0
1	3	2
0	5	90

$= 5$
to the patients
predicted to have
cancer
total predicted
+ve

→ FN is much more dangerous than FP

→ Accuracy = 93%

total no. of
patients with
cancer

Actual: - 1 = 8
0 = 92 } Imbalanced Data

Total actual +ve

Other more important metrics to evaluate class models:-

(i) Precision

$$\frac{TP}{TP + FP} = \frac{3}{5} = 60\%$$

$$= \frac{\# \text{ correct +ve pred}}{\text{Total +ve pred.}}$$

(ii) Recall

$$\frac{TP}{TP + FN} = \frac{3}{8} = 37.5\%$$

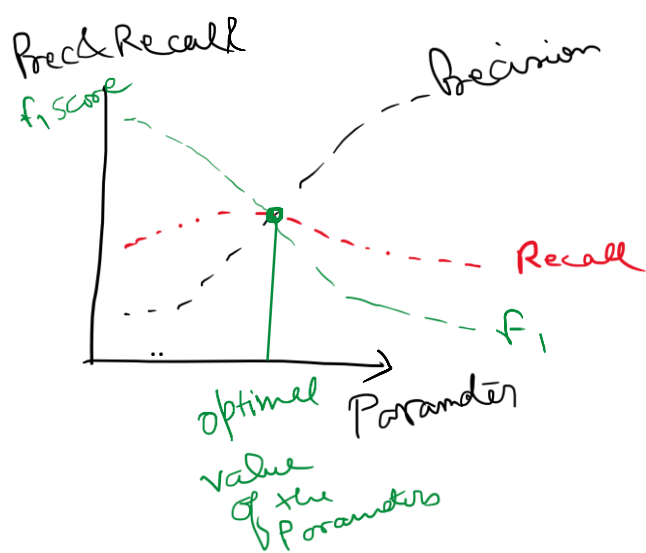
$$= \frac{\# \text{ correct +ve pred}}{\text{Total no. of +ve cases}}$$

→ Higher the values of Precision & Recall, the better the model is.

F₁ - Score :- Harmonic Mean of Precision & Recall

$$= \frac{2 \text{ Prec} \times \text{Recall}}{\text{Prec} + \text{Recall}}$$

All classification models will have an iterable parameter over which Prec & Recall will vary



take the value of the parameters for which F1 score is the highest

HW What is an AUC-Roc curve? Why is it important?

$$\underline{\underline{FPR}} = \frac{FP}{FP + TN}$$

$$\underline{\underline{FNR}} = \frac{FN}{FN + TP}$$

Overfitting

- Increasing training size
- Simplify polynomial order
- Ridge & Lasso (spl. case)
- Early stopping *
- Cross validation

Underfitting

- reduction of noise
- increasing no. of features
- increase polynomial order
- cross-validation
- adjusting hyperparameters

→ HW. What is the difference b/w parameters & hyperparameters?

		Actual	
		Y	N
Pred	Y	TP	FP
	N	FN	TN

Will loan be recovered?

Y
N

FP is worse

Logistic Regression

X y Rent an e-bike or not?

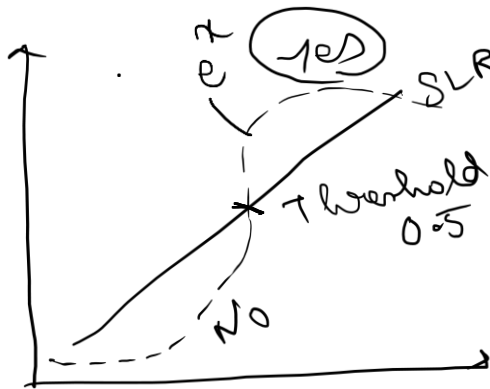
Age	Rent
30	No
49	Yes
45	Yes
28	No
47	Yes
32	No
34	No
38	Yes
42	No
55	Yes

$$y_i = \beta_0 + \beta_1 X_i + \epsilon_i \quad \leftarrow \epsilon_i \text{ needs to be reduced.}$$

SLR cannot partition b/w target variables

$$\pi = P(Y | X=x_i) = \text{Cond. Prob. of } Y \text{ when } X=x_i$$

$$\begin{aligned} \pi &= f(\beta_0 + \beta_1 X) \\ &= \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}} \end{aligned}$$



$$\frac{\pi}{1-\pi} = e^{\beta_0 + \beta_1 X} \quad \xrightarrow{\text{LHS - log odds Ratio}} \log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 X \quad \leftarrow \text{Logit Model}$$

Estimated β_0 & β_1 :- Maximum Likelihood Estimation (MLE)

AIC - Akaike Information Criterion

→ Model Selection technique

(1) Goodness of fit

↓
How well the model explains the data

$$AIC = 2k - 2 \ln(L)$$

↓
Parameters

(2) Model complexity
↓
parameters

Max likelihood func of the model

Lower AIC



Better Model

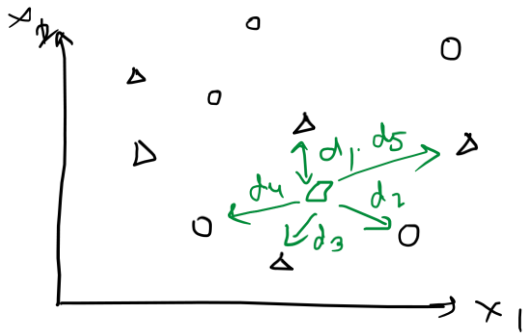
→ Not measure the accuracy of the model

→ Penalize the more complex models.

KNN

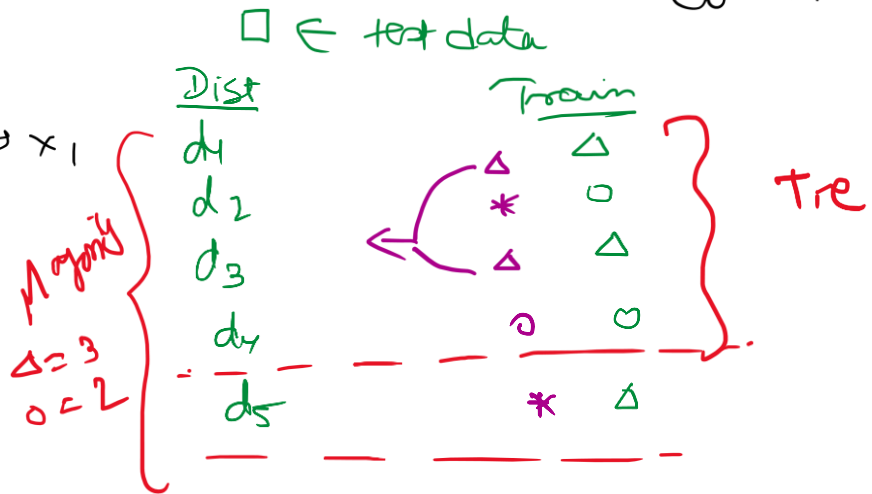
10 obs

x_1 x_2 y



$K = \text{No. of KNN we want to consider.}$

$k=2$



So the predicted dot is Δ.

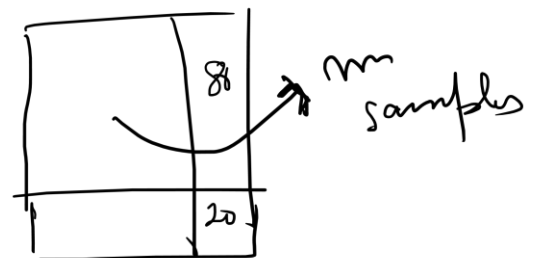
How to choose k ?

① k should be an odd no. [only when it is a binary classification]

② Start with $k=3$ and iterate till $k=\sqrt{n}$
 Eg $n=400$ $k \in [3, 4, 5, \dots, 20]$ $k_{opt} = k |_{\text{max}+10}$

③ Cross Validation - Using diff subsamples of the training data

	$k=1$	2	3	7
$n=1$	✓	~	~	
2				
3				
17				

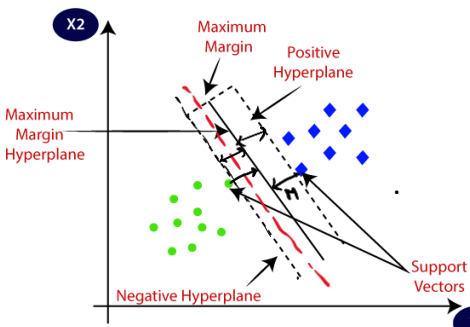


each cell → the performance metric of the value of k .

SVM Support Vector Machine — Eager Learner

Parameters — Internal variables — Estimated (Training Data)
(Coeff. in LR)
(Weights in DT, Log. Reg)
— Automatically updated — Minimize the loss func. using Grad. descent (OLS)
Slope of the tangent at any point in the curve (Can also be named for Grad. desc)

Hyperparameters — External variables → Pre-set → Control the training process and the arch. const. of the model
(Regularizations L_1, L_2)
(Kernel in SVM)
(No. of branches in DT)
→ Not Estimated rather set manually or by automated tuning.

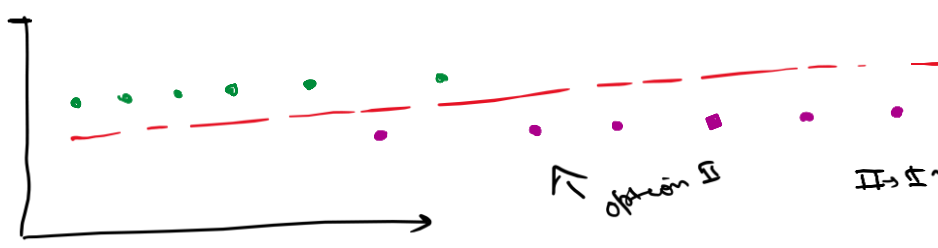


SVM will try to fit a hyperplane that separates 2 class as much as possible.
Maximum margin hyperplane is fitted such that M on either side is equal.
↓ margin

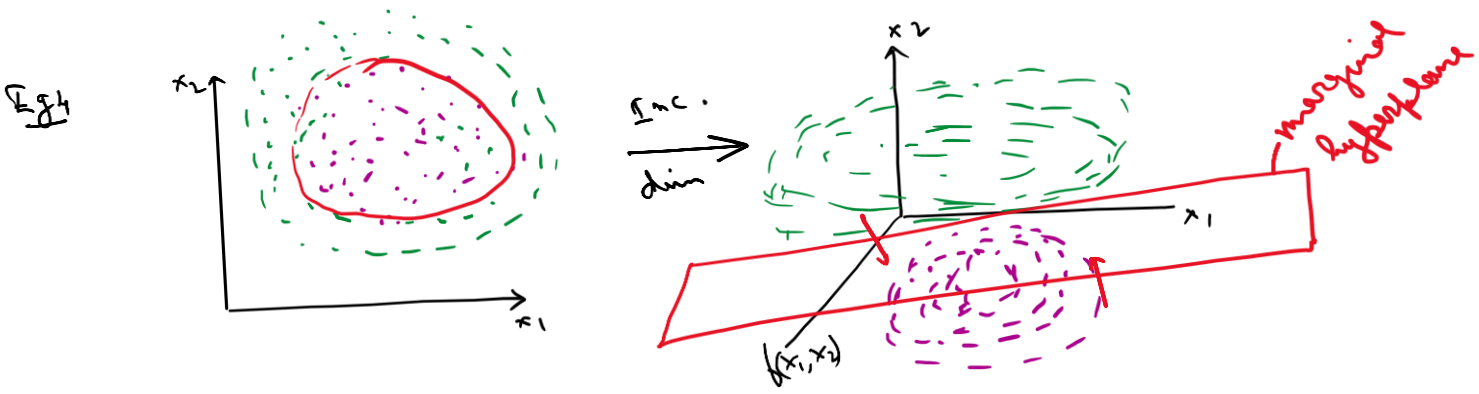
Principle: Higher the value of m , better is the classification



→ 2 options of classification
I → Introduce bias to decrease var., reduce misclassification, increase precision



It's II to introduce a new dimension or plane as the function of the independent variables



- Black box model
- Cannot retraced model path
- SVM is very resource intensive

Understanding the Parameters of SVM

① Kernel type $\rightarrow$ SVM can perform iterative fitting based on kernel type

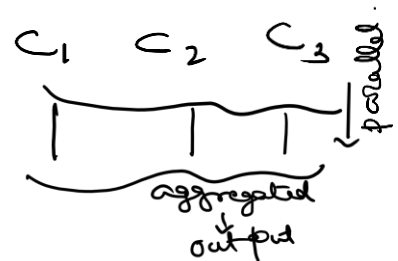
Kernel types $\rightarrow$ Linear
 $\rightarrow$ radial function - rbf
 $\rightarrow$ polynomial $\rightarrow$
 $\rightarrow$ 'pre computed' function

② Penalty parameter $\rightarrow C > 0$
 $\rightarrow$ lower the value of C , lesser is the introduced bias

③ Gamma Regularization $\rightarrow \gamma \in (0, \frac{1}{n})$
 $\rightarrow$ n is the no. of obs. in training data
 $\rightarrow$ low $\rightarrow$ less surrounding data pts will be considered to find the max. margin hyperplane.
 $\rightarrow$ high $\rightarrow$ takes larger surrounding data pts.
 $\rightarrow$ iterating over the entire range.

$\rightarrow$ SVM's are computationally expensive
 $\rightarrow$ Dynamic & parallel computing
 $\rightarrow$ Increase your resource

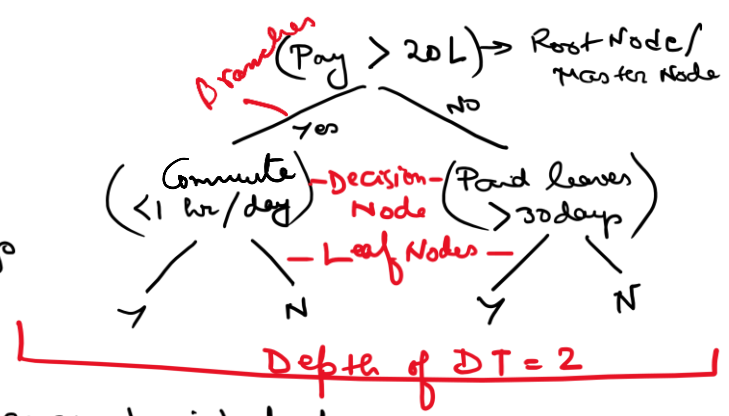
$\rightarrow$ Big data $\rightarrow$ Partition clusters
 $\frac{3PB}{\text{Samples (150 TB) each.}}$



In SVM it is impossible to retrace the path chosen by the model to fit the max. margin hyperplane. So we cannot interpret the model parameters.

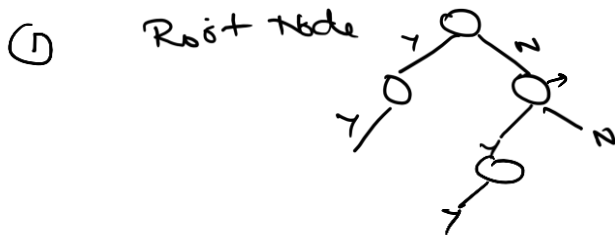
Decision Trees

- Whether to accept a job offer or not?
- Pay > 20 Lakh
- Commute to work < 1 hr per day
- Do they give paid leaves > 30 days



Q) Key advantage of using DT? → Very easy to interpret

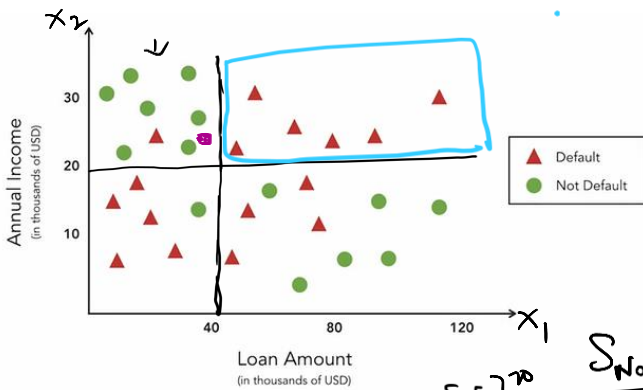
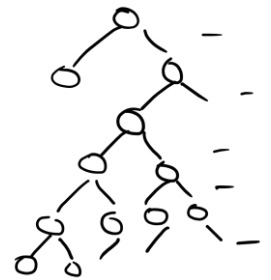
→ Factors that determine the no. of partitions the algorithm will make?



② X_1, X_2, X_3, X_4
 $(X_1, X_2), (X_1, X_3)$
 $(X_3, X_4), (X_2, X_3, X_4)$

All combinations are exhausted
 ↓
 Partitioning will continue

③ Specified Constraints
 max-depth = 5



Total Data pts = 30
 $\Delta = 16$
 $0 = 14$

Steps:

① Create a partition using any one variable so that Impurity on either side of the partition is least

After partition (S_1)

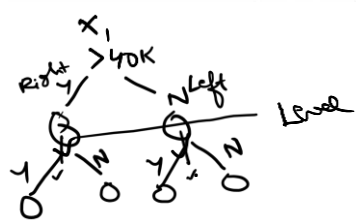
$$2 \begin{cases} S_{left} = \Delta = \frac{6}{14} \\ 0 = \frac{8}{14} \end{cases}$$

After partition (S_2)

$$S_{UL} = \Delta = \frac{1}{8} \\ 0 = \frac{7}{8}$$

$$2 \begin{cases} S_{right} = \Delta = \frac{10}{16} \\ 0 = \frac{6}{16} \end{cases}$$

$$S_{DL} = \Delta = \frac{5}{6} \\ 0 = \frac{1}{6}$$



Annual Inc > 20

$$2 \begin{cases} S_{No partition} \\ \Delta = \frac{16}{30} \\ 0 = \frac{14}{30} \end{cases}$$

Q) Has Impurity dropped after partition? How is Impurity calculated?

$$\rightarrow Entropy(S) = \sum_{i=1}^C p_i \lg(p_i)$$

C = No. of unique classes

p_i = proportion of classes in each side

I Entropy_{NP} = 0.997

II Any partition S_1 has been considered

Left = 0.47 ($X_1 = 40$ in S_1)
 Right = 0.53

$$\begin{aligned} Entropy(S_{left}) &= 0.986 \\ Entropy(S_{right}) &= 0.961 \end{aligned}$$

$$Net Entropy(S_1) = \sum_{i=1}^n w_i Ent S_i = 0.967$$

n = No. of partitions

III Info Gain(S) = Entropy(S_1) - Entropy(No partition)
 = 0.997 - 0.967 = 0.03

Partition is selected in such a way, so that Info Gain(S) is maximum.

Some iterative steps will be conducted for next variables.
 $\rightarrow$ Info Gain due to S_2 must be max.

How to classify a new observation?

$$\frac{X_1}{35} \quad \frac{X_2}{24} \quad \frac{Y}{?}$$

Which Node to select?
 $X_1 < 40$ & $X_2 > 20$

$$\square \rightarrow \begin{matrix} 0 \\ 7 \\ 8 \end{matrix} \quad \begin{matrix} \Delta \\ 1 \\ 8 \end{matrix} \quad \text{So the new obs. is classified to the category with highest freq}$$

For DT Regressions:-

$$(X_1 < 40 \text{ \& } X_2 < 20)$$

$$(4.5, 5.7, 7.8, 6.7, 3.4, 9.1)$$

agg avg

$$\frac{X_1}{30} \quad \frac{X_2}{15} \quad \frac{\text{Continuous Date}}{9.1}$$

Overfitting in Decision Trees

$$\begin{matrix} X_1 & X_2 & X_3 & \dots & X_P & \frac{Y}{A, B} \end{matrix}$$

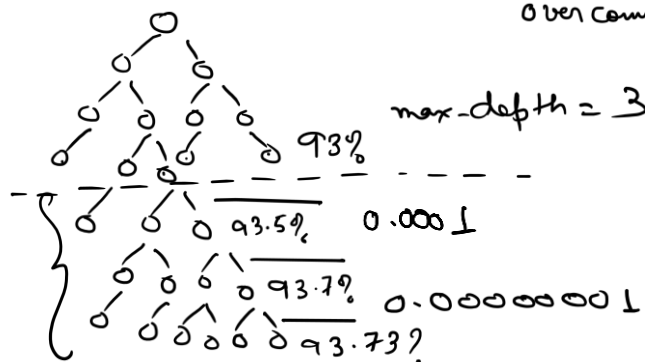
$$\begin{matrix} 1 \\ 2 \\ 3 \\ \vdots \\ n \end{matrix}$$

n - data pt.
 p - indpt var.

$\rightarrow$ Prone to overfitting when $n < p$

$\rightarrow$ Soln ① Increase 'n'

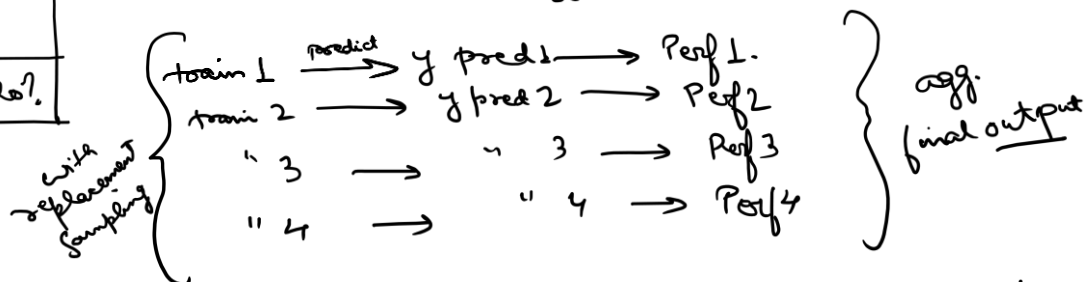
② Use max-depth to restrict over complication



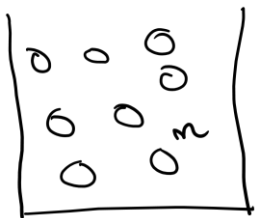
Rule of Thumb: Use DT only when $n > 30$

x	y
train	80%
test	20%

Bagging: Take sub samples of X -train to create multiple DT $\rightarrow$ agg the decision



Bagging in a DT reduces both underfitting & overfitting as much as possible.



n balls in the bag are numbered as

① ② ③ ④ ②

I → Create a sample of 20 balls → Pull out 20 balls at random from the bag.

↳ (n-20)

→ Create a 2nd sample of 20 balls from whatever is left

↳ 20 balls is diff from the 1st sample

↳ (n-40)

Without replacement

II Create a sample of 20 balls → I check them

1 2 4 8 . . . n ← Put them back in the bag

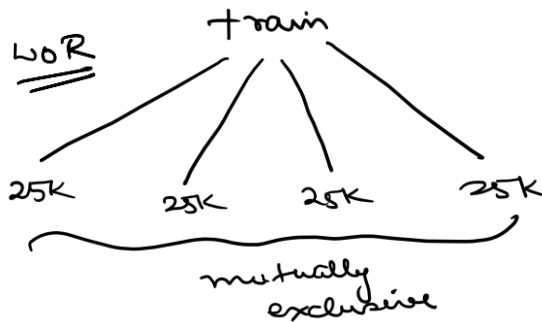
With replacement

Create a 2nd sample of 20 balls → Some of the numbers will repeat from the 1st sample

4 8 10 13 . . . n ← Put them back

Ensemble Techniques

→ All these branches or sample may will have attached bias in case of any seasonal, or underlying trend

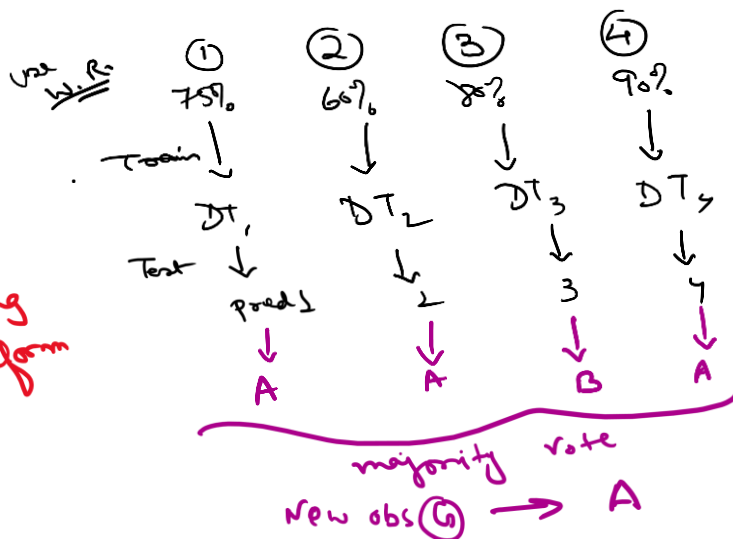


Eg:- 1 Lac data pts

↓ Structured
↓ Does not have any underlying trend

Let's consider a new obs. ⑥.

Boosting
↓
random sub sampling
from a data to perform any opr.



Bagging → Bootstrapping
→ Bootstrapped nodes
→ Aggregating

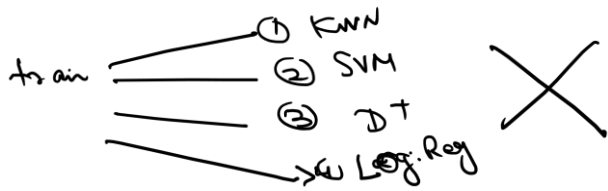
Q) If an ML model is not affected by Missing Values / outliers, do we need to remove or treat them before modelling?

→ No.

→ only DT is not affected by MV/outliers. but other models are.

→ So treatment / removal of MV/outliers is compulsory

Bagging Should be done with same classification model fitted on all subsamples. → Different hyperparameters for different subsamples may be used

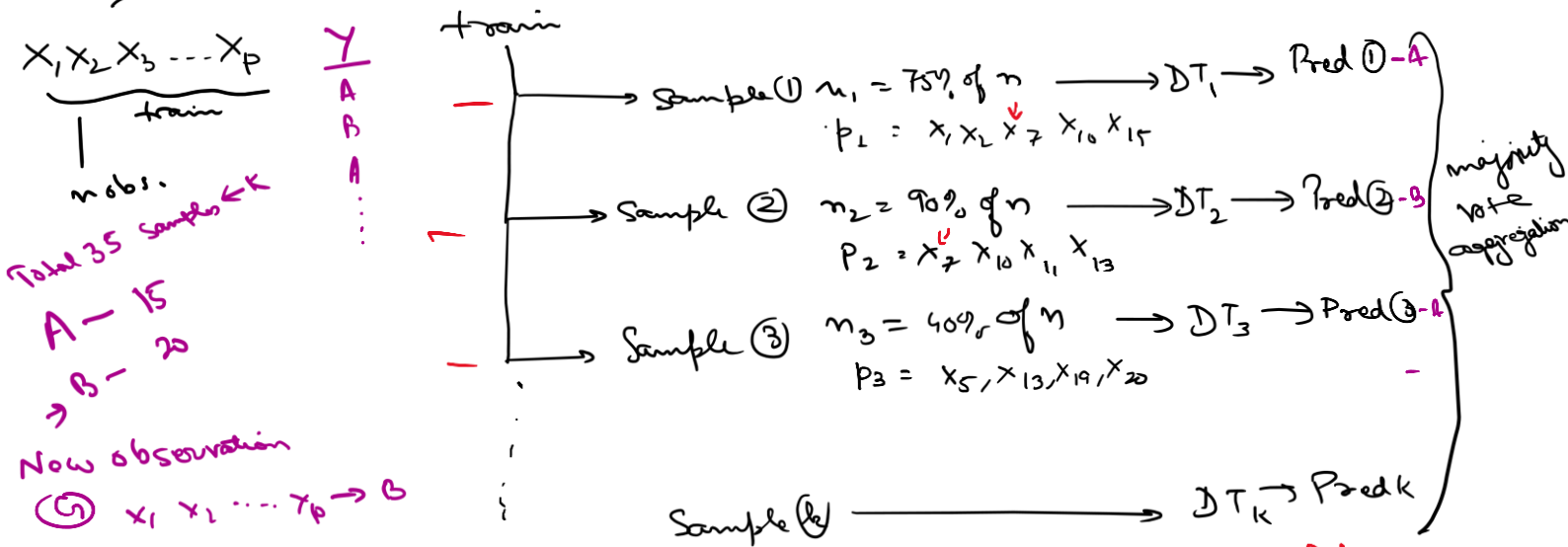


aggregation too we do will not be homogeneous.

What will be the final model be?

Q) What happens if each sub sample will also have a sample of indpt var?

→ Random Forest



→ In Python ^{hyper} parameters will be same by default → In order to have a ^{hyper} parameter Comb. → Specify that.

→ Multiple RF fitting across range of ^{hyper} parameter rather than using diff. parameter values in same RF ^{hyper}

Boosting

Data $x_1, x_2, x_3, \dots, x_p$ y
 $\vdots$ $\in (A, B)$
 n obs.

train a sequence of models.

Sample base

n_1 obs.
 P features

train $\downarrow$

DT_{base}

test $\downarrow$

pred base

Correct pred	Wrong pred
e_1	e_1'
e_2	e_2'
e_g	e_g'

$\downarrow$ $x_1, x_2, x_3, \dots$
 assigned weights

Sample ②

n_2 obs.

P features

train $\downarrow$

DT_2

test $\downarrow$

pred ②

Correct pred	Wrong pred
e_1'	e_1''
e_2'	e_2''
e_g'	e_g''

$\downarrow$ weights of the errors

Sample ③

n_3 obs.

P features

train $\downarrow$

DT_3

test $\downarrow$

pred ③

C	W
e_1''	e_1'''
e_2''	e_2'''
e_g''	e_g'''

$\downarrow$ weights of the errors

Sample ④

How long does it continue / when to stop this iteration of the seq.?

- ① Performance of the subsequent models have saturated
- ② Given threshold of the model performance is achieved

Calculation of assigned weights on misclassification of one sub-model in boosting

Naive Bayes

Obs	X ₁ (Weather)	Y (Play football or not)
1	Sunny	Yes
2	Sunny	Yes
3	Rainy	No
4	Rainy	Yes
5	Cloudy	No
6	Sunny	Yes
7	Rainy	No
8	Cloudy	No
9	Sunny	Yes
10	Rainy	Yes
11	Cloudy	? - Yes

Problem to solve :-

* Given the weather condition as 'Cloudy' to predict if the person will play or not

Cond. Prob

$$P(Y=Yes | X=Cloudy) = \frac{P(Y \cap X)}{P(X)}$$

$$= \frac{2/10}{3/10} = \frac{2}{3} \checkmark$$

Bayes' Theorem

$$P(Y|X) = \frac{P(Y) \cdot P(X|Y)}{P(X)}$$

And prob. of target can be represented as the func. of the cond. prob. of the features

$$P(Y|Cloudy) = \frac{P(Y) \cdot P(X=Cloudy|Y)}{P(Cloudy)} \quad \text{training}$$

model output

Prior prob. $\rightarrow P(X), P(Y)$ because they can be computed independent of each other.

Posterior prob. $\rightarrow P(X|Y)$ because it calculates the occurrence of X based on Y

$$P(X=Cloudy|Yes) = \frac{P(X \cap Y)}{P(Yes)}$$

$$= \frac{2/10}{6/10} = \frac{1}{3}$$

$$\begin{aligned} & \frac{P(Y=Yes | X=Cloudy)}{P(Y=No | X=Cloudy)} \\ &= \frac{2/10 \cdot \frac{1}{3}}{\frac{3}{10}} \\ &= \frac{2}{3} \end{aligned}$$

$$\begin{aligned} & P(Y=No | X=Cloudy) \\ &= \frac{4/10 \cdot \frac{1}{4}}{\frac{3}{10}} \\ &= \frac{1}{3} \end{aligned}$$

$$P(X=Cloudy | No) = \frac{P(X \cap Y)}{P(No)} = \frac{1/10}{4/10} = \frac{1}{4}$$

Since for $X=Cloudy$,
 $P(Yes|Cloudy) > P(No|Cloudy)$
 $Y_{pred} = 'Yes'$

X₁ (Weather)
Categorical

X₂ (No. of players)
Discrete

X₃ (Age)
Continuous

Y (Play or not)
(Yes, No)

$$P(Y | X_1, X_2, X_3) = \frac{P(Y) \cdot P(X_1|Y) \cdot P(X_2|Y) \cdot P(X_3|Y)}{P(X_1) P(X_2) P(X_3)}$$

Bayes' for p predictors :-

$$P(y | x_1, x_2, \dots, x_p) = \frac{P(y) \cdot \prod_{i=1}^p (P(x_i | y))}{\prod_{i=1}^p P(x_i)}$$

CLT
Any distr
↓
Normal
as $n \rightarrow \infty$
74 87
60H 40T
550H 450T
50KH 50KT

Violations in assumptions present :-

① If $X_i \propto X_j$ $i \neq j \Rightarrow$ Drop either X_i or X_j

② Continuous features not normally distributed

Age	Age (11-15, 16-20, 21-25, 26-30)
18	B
25	C
29 $\Rightarrow$	C
13	A
12	A
11	A
19	B
27	D
29	D

① Increase # observations

② Transform the entire data so that X_i is Normal

③ Outlier removal / treatment to check normality

④ Convert the var into robust categories

Python: 3 algorithmic parameters for NB

I> Bernoulli - for binary / multiclass Categorical target

II> Multinomial - for discrete targets / Contⁿ targets convertible to categories

III> Gaussian - for continuous targets

* Gaussian NB can be used accurately for any type of target

Ex

Person developed an web-app.

↳ Improve the performance

- ↳ Page load time
- ↳ Interactive features in the app
- ↳ No. of components
- ↳ Server Bandwidth
- ↳ Image Compression

- X_1 (Avg # clicks in the page) - Discrete

- X_2 (No. of components) - Discrete

- X_3 (# Interactive features) - Discrete

- X_4 (Cache Images present or not) - Categorical

* - X_5 (Cache allocation) - Continuous

Y (Perf.) $\rightarrow$ Continuous

Which model to fit?

i) y - Continuous — Regression / clustering / Robust Categories for classification

0-10
10-20
20-30

$P(Y_{20-30} | X_1, X_2, \dots, X_5)$

!!
?

What is K fold Cross-validation?

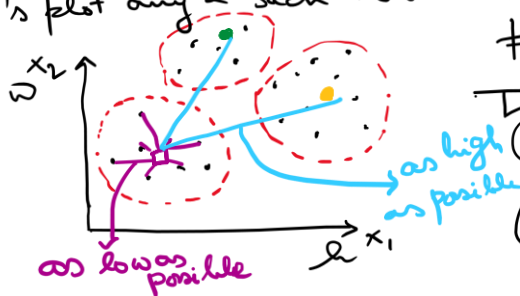
What is Grid Search / random Search / Bayesian Search?

Clustering

eg (Height) (Weight) (Age)

(Income/Income group)

Let's plot any 2 such variables:-



Clusters are not allotted based on visual inspection

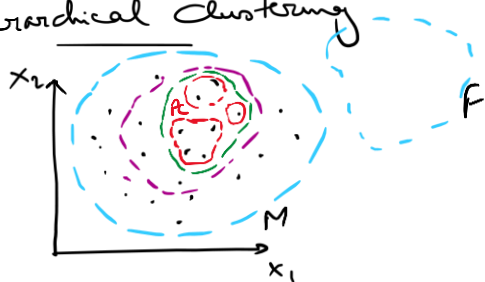
There are 2 objectives:-

- (i) High intra class similarity - within cluster variance very low/as low as possible
- (ii) High inter class dissimilarity - Between clusters variance very high

If there are 3 clusters A, B, C
Dist b/w

A & (B, C)
B & (A, C)
C & (A, B) } these are as high as possible

Hierarchical Clustering



C1 - Male

C2 - Postgrad Males

C3 - Income > 50K with post grad and Male

C4 - Paying rent, Income > 50K, post grad, Male

Each cluster is created so that within-parent cluster segregation of 2 clusters is as high as possible

Parent	Child
C1	> C2, C3, C4
C2	> C3, C4
C3	> C4

} → Child clusters

Based on the algorithm 2 partitioned clustering are:-

- i) K-Means Clustering
- ii) DBSCAN -

X1, X2 have n obs.

Algo:- Step 1:- Choose how many clusters we want (K)
Step 2:- Assign 'k' random pts as starting cluster centers

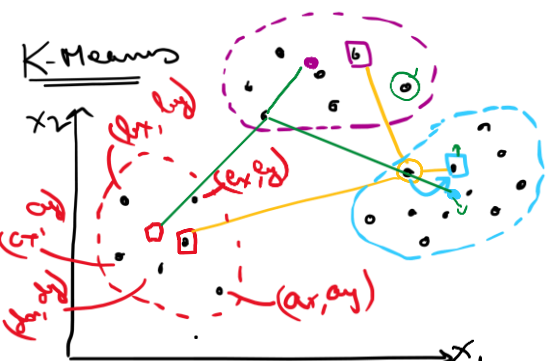
Step 3:- Distance of every obs. is calculated from the current cluster centers

Step 4:- Assign each obs to the cluster of their closest cluster center. So that now we have new clusters

Step 5:- Calculate new cluster centroids and use them as new cluster centers

Iteration of 3-5 times

K-Means



$$C_1 = \frac{a_1 + b_1 + c_1 + d_1 + e_1}{5}, \frac{a_2 + b_2 + c_2 + d_2 + e_2}{5}$$

Q) How to choose k ?

→ We will go on iterating (say 3, 4, 5) until clusters are not shifting location
 → we choose the k clusters obtained as the final clusters.



→ From domain knowledge

→ If k is completely unknown, $k = \sqrt{n/2}$, is a good starting pt. when n is small
 $n = 10,000$ $k = \sqrt{10000/2} \approx 70$ (too excessive).

→ Use 3 different metrics as a judge of optimum k :-

I Elbow method

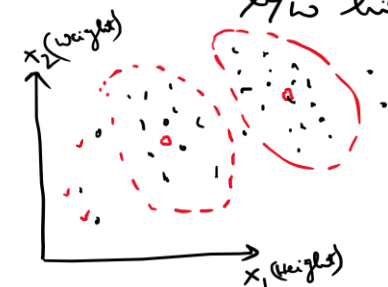
II Silhouette Score method

III C-H Score

DBSCAN

→ K-Means can only create hyper spherical clusters.

→ does not recognize density based clusters i.e., cannot distinguish b/w high vs low density group of data.



→ KMeans will not work with spatially dense data



Some Key Concepts :-

- * (i) ϵ → fixed distance from any obs. in the data space
- (ii) Neighbourhood → data space around an obs. with radius $\epsilon \Rightarrow N_p(\epsilon)$
- (iii)* Min-pt → declared no. of obs. present within neighbourhood of a point
- (iv) Corepoint → an obs. is a corepoint iff within the neighbourhood of the point, there are at least 'min-pt's' no. of observation.
- (v) Borderpoints → " " " " borderpt. iff within the neighbourhood of the point, there are less than 'min-pt's' no. of observation.

(vi) Outlier

* is a hyper parameter in DBSCAN

→ ϵ - Parameter to be specified in the neighbourhood

→ min-pt - reqd to qualify as corepts.

Eg :- min-pt = 4



Core pt.
 $N_p(\epsilon) \geq \text{minpt}$



Border pt.
 $N_p(\epsilon) < \text{minpt}$



Cluster but outlier

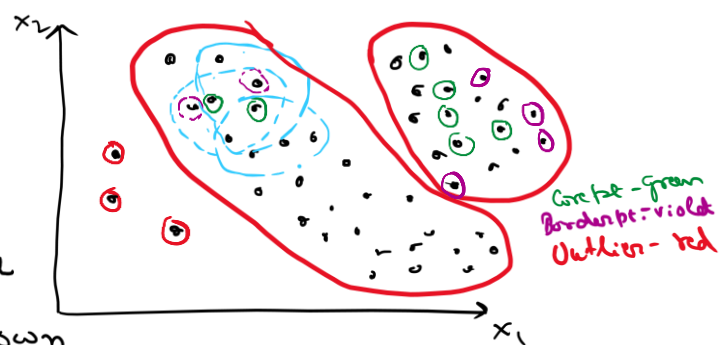
All outliers are also Border points

High density pts are part of a cluster $\text{min-pts} = 4$
v/s

Low density pts are cluster less.

→ Fix ϵ - min neighbourhood distance
min-pts - minimum no. of data pts ^{in the neighbourhood} to qualify as core pt.

→ Each and every pt. is considered to create their own neighbourhood, and checked for core pt. or border pt.



Core pts will induct all other pts in their neighbourhood as a part of their cluster

Border pts will try to be inducted as a part of the cluster if there is a core pt. in their neighbourhood, but will not be able to induct any other pt into their cluster.

Outliers are pts that has no other obs. in their neighbourhood.

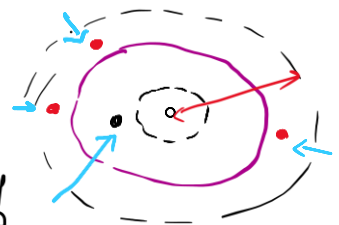
DBSCAN is also used to detect outliers.

→ DBSCAN doesn't require outlier detection before fitting

Q) How to choose optimum value of ϵ & min-pts?

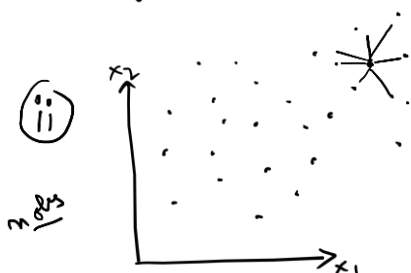
If ϵ is high → too less clusters → underfitting → no detection of outliers

ϵ is low → Small neighbourhoods → overfitting → unnecessary outliers



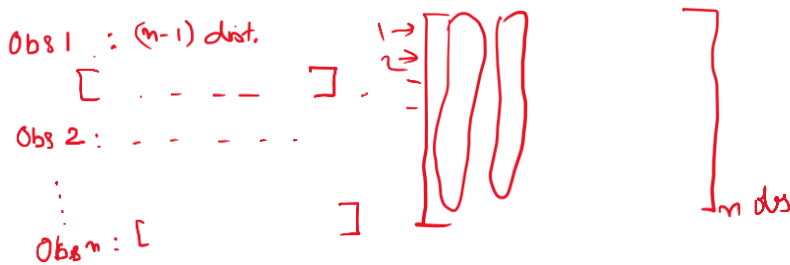
Deciding optimum ϵ :-

(i) After scaling all continuous variables; ran iteration over a range of ϵ and also min-pts

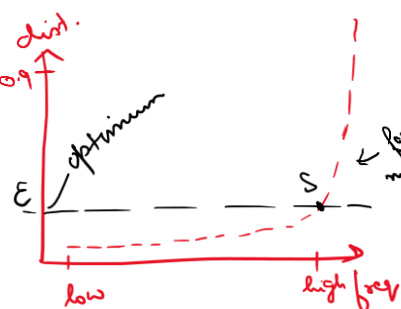


For each obs, we will find the dist. from all other data pts.

Obs 1 : (n-1) dist.
Obs 2 :
⋮
Obs n : []



→ 0.01
b/w
→ 0.02
b/w



→ avg. distances need to be obtained
→ creating a freq. table.

→ Sort the distance of every pt. with other pts in the space and plot them.

→ With increases in distance, freq. of other pts increase, i.e., density reduces.

→ $\text{dist} < S$ → over clustering / overfitting

& $\text{dist} > S$ → under clustering / underfitting

Association Rules/Market Basket Analysis → Figure out which items go together.

WALMART, TARGET

← Departmental Stores

← They sell almost all kind of products

* Quantity is not a concern/not considered

Transaction	Items bought
1	{ bread, milk, eggs } -
2	{ batteries, remote, bread }
3	{ milk, bread, chocolate } -
4	{ chips, biscuits, milk, egg, tomato } -

Objective: If a person buys Item {A}, how likely are they going to buy Item {B}?

① Itemset → Set of unique items in a transaction

② Support {Item} = $\frac{\text{Freq. \{Item\}}}{\text{Total Transactions}}$, for eg:- $\text{Support \{milk\}} = \frac{3}{4} = 75\%$

③ Confidence {X, Y} = If X is bought, probability of Y being bought together with X.

$$= \frac{\text{Support of (X, Y)}}{\text{Support of X}} \quad \left| \begin{array}{l} P(A|B) \\ ? \end{array} \right.$$

④ Lift {X, Y} = $\frac{\text{Support (X, Y)}}{\text{Support(X)} \cdot \text{Support(Y)}}$

$$\frac{\text{Conf(X,Y)}}{\frac{\text{P(X,Y)}}{\text{P(X)} \cdot \text{P(Y)}}}$$

Putting context to a relationship b/w 2 items always depends on the analyst's perspective
 Rule Set/Association Rule: $\{X\} \rightarrow \{Y\}$ If X is bought then Y is bought
 antecedent consequent

Ex: {milk} → {eggs} : Z
 Scenario where milk is bought so eggs may be bought!

Lift {Z} → Lift is a metric that can be used to evaluate the strength of the association rule. → Zhang's metric can also be used

Applications of Association rules:-

- Healthcare - Drugs and their efficacy against diseases
- Social Media Posts
- Stock Analysis
- Food order

Frequent ✓
 A: {bread, milk} → eggs
 Support(A) > threshold
 frequent

Infrequent - Itemset
 B: {batteries} → {bread}
 Support(B) < threshold
 Infrequent

Q) How to determine threshold?
 → Threshold is the only hyper-parameter for Association Rule algo.

Using this Apriori Method to find Association Rules:
 → We get 3 types of insights:

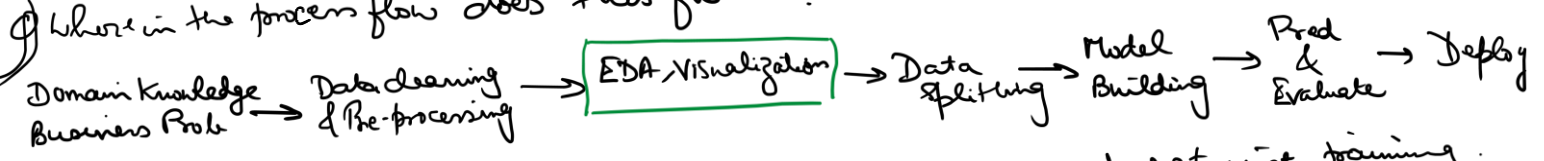
Obvious
 rational context
 and can be easily
 identified
 {bread, milk} → {eggs}

Abstruse
 No rational context
 can be easily identified
 {Diapers} → {beer}

Actionable
 rational context,
 is developed by domain
 or in practicality/action
 → {sock, shoes} → {umbrella}
 [Context: It is the rainy season
 and schools, colleges & offices
 have started]

Day 16 Dimension Reduction — Feature selection → Select only non redundant & high var features
 Feature Extraction → Extract a subset of features or a comb. of these to create a new feature which best represents the original set!
 → Feature Engineering: — We have 10 features $X_1, X_2, X_3, \dots, X_{10}$ $p \geq 10$
 → The idea of dimension reduction is to use $k < p$ features but not lose data/information as far as possible.

Where in the process flow does this fit in?



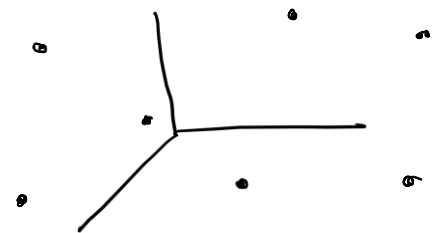
→ Dimensionality Reduction is performed on the entire data and not just training.
 Hence it is done on clean & preprocessed data to make it ready for modelling.

Curse of Dimensionality: Data set has p features ($p = 150$)

$X_1 \quad X_2 \quad \dots \quad X_p$
 $\rightarrow 3.1 \quad 4.2 \quad \dots \quad 7.1$

→ When working with high dim data, data tends to become sparse as the volume of the space increases exponentially

→ Distance metrics become meaningless
 → overfitting



Smallest gap $\approx$ largest gap

Soln: ① Dimensionality Reduction — PCA
 ② Feature Selection —
 ③ Regularization: $L1/L2$

Decrease model complexity as well as tackle overfitting

Regression: Feature selection methods:-

① Forward Selection:-

[Data: p features & n obs. ; $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$]

Start with a null model:-

$$y = \beta_0 \rightarrow \text{null model}$$

p possibilities
are there

$$\text{I } y = \beta_0 + \beta_1 x_1 \rightarrow \text{Adj } R^2_{\text{I}}$$

$$\text{II } y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \rightarrow \text{Adj } R^2_{\text{II}}$$

⋮

In total
there are

$p!$ possibilities

$$\phi y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p \rightarrow \text{Adj } R^2_{\text{p}}$$

If $\text{Adj } R^2$ of any step is greater than
prev. step then the added feature in
that step is considered important.
Else, the added feature is non
relevant / redundant
↳ choose to drop it

Backward Selection:

Starting with full model and proceed to the null model

$$\text{I : } y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p \quad \left. \begin{array}{l} \text{Compare the perf.} \\ \text{using Adj } R^2, \text{ MSE} \end{array} \right\}$$

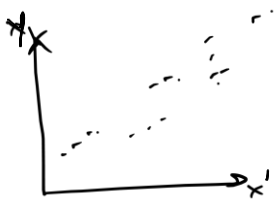
$$\text{II } y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_{p-1} x_{p-1}$$

⋮

If we see that $\text{Adj } R^2$ is increasing
or remains same, that var. was
redundant,
Else, we have to drop some other var
and check.

In clustering problem, we can use this on a subset of features to get the
feature importance, so that we can exclude redundant variables and
use only relevant var for the clustering algo.

Feature Extraction: Factor Analysis



$$\begin{matrix} x_1 & x_2 & x_3 & x_4 \\ \vdots & \vdots & \vdots & \vdots \\ f_1 \begin{bmatrix} \cdot \\ \cdot \\ \cdot \end{bmatrix} & f_2 \begin{bmatrix} \cdot \\ \cdot \\ \cdot \end{bmatrix} & f_3 \begin{bmatrix} \cdot \\ \cdot \\ \cdot \end{bmatrix} & f_4 \begin{bmatrix} \cdot \\ \cdot \\ \cdot \end{bmatrix} \end{matrix} \quad \text{factors}$$

$$\text{We get a new feature } X' = L_1' f_1 + L_2' f_2 + L_3' f_3 + L_4' f_4$$

$$X' = L_1 x_1 + L_2 x_2 + L_3 x_3 + L_4 x_4$$